



THE REPUBLIC OF LIBERIA
LIBERIA MARITIME AUTHORITY

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Marine Advisory: 01/2016

SUBJECT: Carriage of Bauxite and other solid bulk cargoes that may liquefy

Dear Shipowner/Operator/Master:

On the 2nd January 2015, the bulk carrier BULK JUPITER carrying a cargo of approximately 46,400 tonnes of bauxite on a voyage from Kuantan, Malaysia, to Qingdao, People's Republic of China, sank approximately 150 nautical miles off the coast of Vietnam with the loss of 18 lives from a crew of 19.

The cargo of bauxite was declared as a GROUP C (cargoes neither liable to liquefy –GROUP A or possess chemical hazards- GROUP B) cargo under the IMSBC Code. However, some initial considerations of the circumstances surrounding loss of the BULK JUPITER suggested that a possible causal factor in the casualty was loss of stability resulting from the liquefaction of the cargo on board the ship.

Pending the results of ongoing research into the properties of bauxite and the risk of liquefaction, Shipowners, Operators and Masters should not accept bauxite cargo for loading unless it meets the conditions in the attached IMO circular, CCC.1/Circ.2

Masters are also encouraged to consult section 7 of the IMSBC Code, for additional information on cargoes that may liquefy and may carry out a check test for approximately determining the possibility of flow by the auxiliary method prescribed in section 8 of the IMSBC Code.

For more information please contact the Technical Department at Technical@liscr.com or telephone +1 703 790 3434.

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CCC.1/Circ.2
20 October 2015

CARRIAGE OF BAUXITE THAT MAY LIQUEFY

1 The Sub-Committee on Carriage of Cargoes and Containers (CCC), at its second session (14 to 18 September 2015), considered submissions made by the Bahamas in document CCC 2/5/16 relating to the carriage of this cargo, including some initial considerations of the circumstances surrounding the loss of the 10-year-old Bahamas flag Supramax bulk carrier **BULK JUPITER** with the loss of 18 lives on 2 January 2015.

2 The cargo of bauxite was declared as a Group C cargo under the International Maritime Solid Bulk Cargoes (IMSBC) Code. However, the Sub-Committee noted the information in the Bahamas document CCC 2/5/16 that loss of the vessel may have been caused by liquefaction of the cargo. The Sub-Committee also considered proposals made by Australia and co-sponsors with respect to the properties of bauxite and the need for the cargoes material properties to be further examined in order to ensure that the cargo can be carried safely.

3 Pending the outcome of ongoing research, the Sub-Committee concluded that:

- .1 there is a need to raise awareness, despite the efforts to date made by the Member clubs of the International Group of P&I Clubs, of the possible dangers of liquefaction associated with carriage of bauxite;
- .2 the potential for bauxite to liquefy is not specifically addressed in the IMSBC Code, since it is only classified as Group C cargo;
- .3 if a Group A cargo is shipped with moisture content in excess of its transportable moisture limit (TML) there is a risk of cargo shift, which may result in capsizing;
- .4 the master should not accept this cargo for loading unless:
 - .1 the moisture content of the cargo indicated in the certificate is less than the indicative moisture limit of 10% and the particle size distribution as is detailed in the individual schedule for BAUXITE in the IMSBC Code; or

- .2 the cargo is declared as Group A and the shipper declares the TML and moisture content in accordance with paragraph 4.3.1 of the IMSBC Code; or
- .3 the competent authority has assessed the cargo and determines that the particular cargo does not present Group A properties. Such assessments shall be provided by the shipper to the master as required by paragraph 1.2.1 of the IMSBC Code;
- .5 if the master has reason to doubt that the cargo being loaded is consistent with the shipper's declaration then the master should stop loading and have the shipper verify the properties of the cargo¹. If necessary, advice should be sought from the competent authority of the country of loading; and
- .6 if the cargo is declared as Group A, the master should refer to section 7 of the IMSBC Code, which warns about cargoes that may liquefy.

4 The International Maritime Organization is taking action to investigate the hazards and risks associated with the carriage of bauxite and, upon finalizing that investigation, would consider any necessary amendments to the IMSBC Code².

5 Member Governments are invited to bring the above information to the attention of shippers, terminal operators, shipowners, ship operators, charterers, shipmasters and all other entities concerned, requesting that extreme care and appropriate action be taken, taking into account the provisions of relevant IMO instruments when handling and carrying bauxite in bulk.

¹ Refer to regulation 2 of chapter VI of SOLAS 1974, as amended.

² The CCC Sub-Committee has established a correspondence group (CG) and Member Governments and international organizations are invited to submit relevant information regarding the safe handling and carriage of this cargo, at their earliest convenience, to the Organization. The outcome of the CG will be considered by the Sub-Committee at its third session in September 2016.